

Caught in the Storm: Extreme Weather Hazards

This video by the USCSB covers the incident at the Arkema Chemical Plant that occurred in 2017 during and after the landfall of Hurricane Harvey in Crosby Texas. The video dives deep into what occurred during the Hurricane, highlighting the bravery of the hurricane crew, the faulty actions of the first responders, and the lack of guidelines and preparations by forces larger than Arkema.

In 2017, Hurricane Harvey made landfall in Texas. Prior to the first rain, the entire plant, which produces organic peroxides, had been shut down and prepared. All peroxides were kept in warehouses and trailers, which were cooled below 32 degrees to prevent the peroxide from decomposing and catching fire, the major risk factor in the product. As the first rain began to fall, the only people at the plant were the “ride out” crew, which were there to oversee the plant during the hurricane. Their actions would significantly alter the course of events over the next 72 hours.

As the rain fell, the water level rose. Since its construction in 1960, the plant had been considered above the flood zone. Only in 2007 did new information reveal that they were not outside the flood zone, almost 50 years after the construction of the plant. Because of this and the reports from local authorities, the ride out crew believed that the water levels would stop rising and that there was no further risk to the product or buildings. Despite their beliefs, the water levels continued to rise. At a certain level, the electrical systems of several warehouses were at risk of flooding, and so the ride out crew moved the product out of those warehouses and to another, which sat on higher ground, shutting off the power to those buildings. Still, the rain kept falling. Eventually, the water levels rose too much, causing the main transformers to flood and stop functioning. The backup generators, which sat at a higher elevation, kicked on, cooling the last warehouse and attached cooling trailers. Fearing that the flood waters would reach the backup generators, the ride out crew moved the remaining product to cooling trailers parked on higher ground, just in time for the backup generators to stop working. During this movement, the truck used to move the trailers flooded, leaving three cooling trailers still attached to the warehouse. As there was still product in the warehouse, workers moved peroxide via forklift to the attached trailers. When the forklift flooded as well, the workers moved the last 2,000 containers by hand through the floodwaters. Despite this, the water kept rising, causing the remaining trailers to fail. The peroxides were no longer cooled, and began to decompose.

The ride out crew was evacuated (in addition to a 1.5 mile radius around the plant, which included around 200 people), and several hours later several of the trailers caught fire, releasing toxic gasses to the surrounding area. Police, traveling near the highway, were exposed to the fumes and reported a white smoke. After this initial report, the highway was closed. Yet after being checked several hours later, there was no smoke detected, and the highway was reopened due to its importance in evacuation procedures for the surrounding area. Smoke was detected again on the highway, causing several people to seek medical attention. No deaths were reported.

This incident was, I believe, to be an *extreme* edge case. Hurricane Harvey delivered so much water it was deemed a “greater than 500 year storm.” The ride out crew was valiant in their efforts to maintain the product in cooled locations, moving product through the night and even

by hand for hours to keep the peroxides from decomposing. There were no operational issues that caused this accident, nor were there maintenance issues that could have been prevented. In fact, there were a large number of backup and failure systems in place, but none of the equipment was prepared for such a high level of water. The biggest thing that could have been done for prevention was accurate analysis of the flood zone and preparation. Even though the plant was deemed to be in a flood zone in 2007, the insurance company and other regulatory bodies had no recommendations for the plant, deeming it “safe” from floods. Clearly, this was not true. Not only were there no recommendations, there were no overarching guidelines for chemical plants of this type for preparation, action, or planning for a 500 year storm such as hurricane Harvey. The biggest takeaway is that companies and regulators need to be extra safe and have a conservative approach when evaluating risk, including ensuring that there is no single failure mode for all systems (though I would argue it is hard for a system to not be susceptible to damage when being completely underwater).

The video had excellent quality, though the repetition of simulated clips was a bit boring. I rate it a 7/10. The effectiveness of the video I rate a 5/10, as I would say the biggest take away from the video, which was mirrored in the Youtube comments, was the bravery and effort of the ride out crew, who did everything in their power to keep the product from combusting. While it calls on regulatory bodies and plant engineers to be more conservative in their flooding estimates, many engineers who are making financial decisions would consider building for a 500 year flood when the plant may only be in operation for 10, 50, or 75 years is a fool’s errand. I still believe the message to be important, hence the higher score.

For a safety exercise, I would love to suggest something similar to Dr. Cooper’s senior design safety video game. I recommend (as I would find it fun and engaging), a type of Sim-City type game, where you physically design a plant (simple layout, with drag and drop building blocks), and oversee the operation over the course of 10, maybe 50 years, speeding up time to quickly pass through the years. Every once in a while, an extreme event would happen, such as the damage of equipment due to not practicing maintenance checks, or major weather events like hurricanes or extreme cold. This would give you a real idea of how often these things are, how to plan/prepare for them, and their consequences. You could grade based on the number of failures, or on how well the plant survived, or simply on a reflection type paper such as this one. I believe it would be very beneficial. Thank you for a wonderful semester!